Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Audio, Speech & Acoustic AI | Deep Learning, Real-Time Inference

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Researcher, Speech & Acoustic AI - Huawei Canada (Markham, ON)

Dear Hiring Manager,

I am applying for the Researcher, Speech & Acoustic AI position at Huawei Canada. I am a machine learning researcher with a PhD from the University of Trento (2025), 10+ years across industry and academia, and a research focus on audio and sensor-stream ML. My most recent published system made real-time decisions in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026).

Here is how my work maps to the role:

- **Acoustic and audio projects.** I built audio pattern-detection pipelines end to end (DSP features, RNN/LSTM models, edge deployment), released four open-access musical audio datasets (9,800+ recordings, 70+ musicians), and wrote the open-source C++17 library Nebula for audio feature extraction with per-feature ARM latency benchmarks.
- **Frontier research to publication.** I design benchmarks with baselines and ablations, most recently 6 methods over 100,000 scored trials for a speech-pronunciation classifier (Wav2Vec2 XLSR-FR reached ROC-AUC 0.822 vs. 0.599 for DTW-MFCC), and I have carried work through to 10+ peer-reviewed papers (JAES, IEEE).
- **Research that lands in products.** On the EU-funded MUSMET project I owned the pipeline from acquisition to deployment, with <30 ms inference at >90% F1, working with academic and industry partners.

To be direct about the gaps: my publications are at audio-engineering and IEEE venues (JAES, IEEE IS2, I3DA), not at ACL, ICLR, AAAI or NeurIPS, and my post-PhD record is a one-year postdoc plus independent research. My strengths are rigorous evaluation, constrained-compute deployment and audio signal processing, and I would pair them with modern speech models.

I am based in Toronto, so the Markham office is within reach. I am a Canadian permanent resident, eligible to work in Canada without sponsorship, and available immediately. I would welcome the chance to discuss how my work on speech, acoustic and audio ML can support the research agenda at Huawei Canada.

Sincerely,

Nishal Stanislaus Silva, PhD